Evaluating RobotReviewer for Automated Risk of Bias Assessment in a Systematic Review: A Case Study

Mary Edwards and Chris Marshall

York Health Economics Consortium, Enterprise House, Innovation Way, University of York, Heslington, York, YO10 5NQ

BACKGROUND & OBJECTIVES

Risk of bias (RoB) assessment is an important part of a systematic review and hence the production of health technology assessment. It is also a time consuming[1], subjective[2] and labour intensive process, and disagreements on a study's RoB between reviewers are common. In response, novel software tools have emerged which aim to support this process (a selection of these can be found catalogued in the Systematic Review Toolbox[3]).

RobotReviewer (RR) is a free web-based machine learning system that aims to automate RoB assessments of randomised controlled trials (RCTs)[4]. RobotReviewer has been tested internally by its developers where it performed well, but to date we have not identified any published independent evaluations.

We compared and evaluated RobotReviewer against the current standard for RoB assessment, defined as double, independent, human researcher assessment with disagreements resolved by a third reviewer.

The objective of the work was to evaluate:

"How effective is RobotReviewer at assessing risk of bias of RCTs as compared with double independent reviewer human assessment, with any disagreements adjudicated by a third (human) reviewer?"

METHODS

RobotReviewer was tested on a set of 35 RCT papers that had been previously assessed by two independent human researchers, as part of a systematic review.

Results of the manual, human reviewer assessments were edited to be comparable with the automated (i.e. RobotReviewer) assessments.

The results of the automated assessments were compared with the manual, human reviewer assessments for similarity at each RoB domain in accordance with the Cochrane Risk of Bias Tool.

Our test set of 35 studies had been assessed on 8 fields with 3 risk options per field. To make this output comparable with the output of RR, we:

- Equated "Unclear" and "High" with the RR single option "High/unclear";
- Conflated fields where necessary, i.e. the separate "participant blinding" and "personnel blinding" fields became one field in line with the version of the Cochrane tool used by RR.

Figure 1: Data Collection

	Study ID	Was there adequate sequence generation?	Was allocation adequately concealed?	Adequate blinding of participants and personnel?	Adequate blinding of outcome assessment?	Were incomplete outcome data adequately addressed?	Are reports of the study free of suggestion of selective outcome reporting?	Overall risk of bias	% Agreement
Human	Mi 2015 {, #8}	Low	Unclear	Low	Low	Low	Unclear		
RR		+	+	?	?				
Agreement		Yes	No	No	No				25%
Human	Savino 2015 {, #9}	Low	High	High	Low	Low	Low		
RR		+	+	?	+				
Agreement		Yes	No	Yes	Yes				75%
Human	Kianifar 2014 {, #10}	Low	Unclear	Low	Low	Low	Low		
RR		+	+	+	?				
Agreement		Yes	No	Yes					50%
Human	Sung 2014 {, #11}	Low	Low	Low	Low	High	Low		
RR		+	+	+	+				
Agreement		Yes	Yes	Yes	Yes				100%
Human	Skjeie 2013 {, #12}	Unclear	Low	Low	Low	High	Unclear		
RR		+	+	+	+				

At the time of testing, RR was not able to make an assessment of two fields in the Cochrane RoB tool (see blue highlight in Figure 1). Therefore, human reviewer and automated assessments have only been compared across four of the six fields.

RESULTS

The mean level of agreement between RobotReviewer and double, independent, human researcher assessment was 72% across all papers. Of the 35 papers:

- 25% agreement was achieved on 3 papers;
- 50% agreement was achieved on 8 papers;
- 75% agreement was achieved on 14 papers;
- 100% agreement was achieved on 10 papers.

Agreement across domains ranged from 69% for the blinding field to 77% for the sequence generation field.

Table 1: % agreement by field

Field	% agreement		
Was there adequate sequence generation?	77%	Mean	72%
Was allocation adequately concealed?	71%		
Adequate blinding of participants and personnel?	71%		
Adequate blinding of outcome assessment?	69%		
Was incomplete outcome data adequately addressed?	0%		
Are reports of the study free of suggestion of selective outcome reporting?	0%		

The software achieves more consistent results across fields than across papers, suggesting that there are some papers that it either a) struggles to accurately assess, or b) consistently assesses differently to the human reviewers. Further work is needed to determine what these papers have in common.

CONCLUSIONS

This study has provided practical insight into the current effectiveness of employing a machine learning system to support RoB assessment in a systematic review, as part of a health technology assessment.

PROS

- Very quick and easy to use;
- Potentially a huge time saver for resource limited reviewers;
- Easy to see which pieces of information have been used to make each judgement.

CONS

- RR doesn't always achieve the same output as human reviewers;
- To anyone who cannot read the code it is not clear how/why RR is making the assessments it does;
- RR can only assess PDFs and full papers (it doesn't perform so well on abstracts);
- Only readable PDFs can be assessed – no image files or Word documents;
- At the time of testing, RR only assessed 4 of the 6 domains in the Cochrane RoB tool.

REFERENCES

1. Tovey *et al.* Fit for purpose: centralising updating support for high-priority Cochrane Reviews. National Institute for Health Research Evaluation, Trials, and Studies Coordinating Centre. 2011. **2.** Hartling *et al*. Applying the risk of bias tool in a systematic review of combination long-acting beta-agonists and inhaled corticosteroids for persistent asthma. *PloS one.* 2011. **3.** Marshall, C. Systematic Review Toolbox (http://systematicreviewtools.com). **4.** Marshall *et al.* RobotReviewer: evaluation of a system for automatically assessing bias in clinical trials. J Am Med Inform Assoc. 2016.

CONTACT US

mary.edwards@york.ac.uk
Telephone: +44 1904 323437
Website: www.yhec.co.uk
MINERVA http://www.minerva-network.com/

http://tinyurl.com/yhec-facebook
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